

Reasoning about Welfare-Affecting Capabilities in Concurrent Games

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Strategic ability is not only about achieving goals

In multi-agent systems, a coalition's choices may also affect the welfare of other agents.

Classical ATL/CL asks

- Can coalition C ensure φ ?
- Can C keep φ true forever?

We also ask

- Can C benefit coalition B ?
- Can C harm coalition H ?
- Can it do so while ensuring φ ?

Who is a potential benefactor, and who is a danger?

Conceptual Ingredients

Harm and benefit

We adopt a consequentialist, non-comparative reading, where good and bad outcomes are represented through agents' preferences:

- harm: bringing about a bad outcome;
- benefit: bringing about a good outcome.

Danger

We follow a modal, non-probabilistic intuition for danger [Fassio, 2025, Garland, 2003, Peschard et al., 2022]:

- a coalition is dangerous if it has the potential to harm;
- potential means existence of a suitable strategy.

strategic capability [Alur et al., 2002, Goranko and van Drimmelen, 2006]

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preference over outcomes [Li et al., 2025]

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welfare-affecting capability

- 1 A semantics based on concurrent games enriched with agents' preferences over computations.
- 2 New ATL and CL modalities for welfare-affecting strategic capabilities:

$$\langle\langle C:B,H \rangle\rangle\gamma$$

“ C has a strategy that benefits B , harms H , and guarantees γ .”

- 3 Technical results:
 - sound and complete axiomatization for the until-free fragment;
 - polynomial embedding into standard ATL/CL with special atoms marking benefited and harmed agents;
 - tight satisfiability complexity: Exptime-complete for WA–ATL, Pspace-complete for WA–CL.

Models: Concurrent Games with Preferences

A model is a concurrent game model with regular preferences:

$$M = (\mathbb{S}, \text{ACT}, (\mathcal{R}_\delta)_{\delta \in \text{JACT}}, \preceq, \mathcal{V}).$$

- $\mathbb{S}$: non-empty set of states.
- ACT : action names; $\text{JACT} = \text{ACT}^n$: action profiles.
- $\mathcal{R}_\delta \subseteq \mathbb{S} \times \mathbb{S}$: transition relation for action profile δ .
- for each agent i and state s , a regular preference order

$$\preceq_{i,s}$$

over computations starting at s .

- $\mathcal{V} : \mathbb{S} \longrightarrow 2^{\mathbb{P}}$: valuation function.

Constraints for game structures

Collective choice determinism, independence of choices, and seriality.

Regularity of preferences

The preference order $\preceq_{i,s}$ is regular if it is a total preorder over computations starting at s and:

- stable:
for all computations λ, λ' starting at s , if $\lambda(1) = \lambda'(1)$ and $\lambda \preceq_{i,s} \lambda'$, then $\lambda_{\geq 1} \preceq_{i,s'} \lambda'_{\geq 1}$, where $s' = \lambda(1)$;
- well-founded:
the worst computation exists;
- conversely well-founded:
the best computation exists;
- non-flat:
there is λ, λ' starting at s such that $\lambda \prec_{i,s} \lambda'$.

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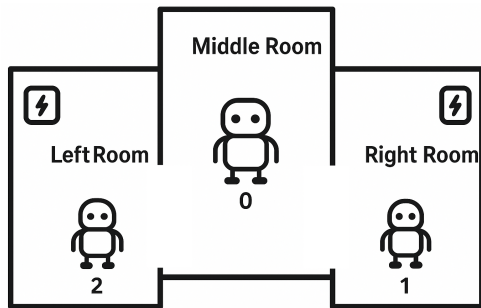
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Running Example: Robots and Charging Rooms



- Robot 0 starts in the middle room.
- It wants to reach a charging station on the left or right.
- Robots 1 and 2 guard the right and left passages.
- Each guard can block or free its passage.

Preference profile

Robot 0 prefers computations where it eventually reaches a charging station.

Robots 1 and 2 prefer computations where robot 0 never reaches one.

Strong Benefit and Strong Harm

Let $\mathcal{O}_F(s, \mathcal{F}_C)$ be the computations compatible with strategy $\mathcal{F}_C$ from s .

Strongly beneficial for B

$$\mathcal{F}_C \in \text{StrSB}_M^C(s, B) \quad \text{iff} \quad \mathcal{O}_F(s, \mathcal{F}_C) \subseteq \text{Best}_{i,s} \text{ for every } i \in B.$$

Strongly harmful for H

$$\mathcal{F}_C \in \text{StrSH}_M^C(s, H) \quad \text{iff} \quad \mathcal{O}_F(s, \mathcal{F}_C) \subseteq \text{Worst}_{i,s} \text{ for every } i \in H.$$

Strong benefit/harm means guaranteeing best/worst outcomes for the affected coalition.

Fact

The coalition of robots $\{1, 2\}$ has a strategy that is harmful to robot 0 and beneficial to the guarding coalition.

$$\begin{aligned}\varphi, \psi &::= p \mid \neg\varphi \mid \varphi \wedge \psi \mid \langle\langle C:B,H \rangle\rangle\gamma \\ \gamma &::= X\varphi \mid G\varphi \mid \varphi U\psi\end{aligned}$$

Key modality

$$\langle\langle C:B,H \rangle\rangle\gamma$$

Coalition C has a strategy which is beneficial for B , harmful to H , and guarantees temporal property γ .

Standard ATL is recovered as the special case:

$$\langle\langle C \rangle\rangle\gamma \equiv \langle\langle C:\emptyset,\emptyset \rangle\rangle\gamma.$$

$$(M, s) \models \langle\langle C:B, H \rangle\rangle \gamma$$

iff there exists a strategy

$$\mathcal{F}_C \in \text{StrSB}_M^C(s, B) \cap \text{StrSH}_M^C(s, H)$$

such that every generated computation satisfies γ :

$$\forall \lambda \in \mathcal{O}_F(s, \mathcal{F}_C), \quad (M, \lambda) \models \gamma.$$

Three roles in one operator

- C : the acting coalition;
- B : the coalition benefited by the strategy;
- H : the coalition harmed by the strategy.

Potential Benefactor and Danger

The language gives compact definitions of the target notions.

Potential benefactor

$$\text{Benef}(C, B) \equiv \langle\langle C:B, \emptyset \rangle\rangle X\top.$$

Coalition C has a strategy beneficial for B .

Danger

$$\text{Danger}(C, H) \equiv \langle\langle C:\emptyset, H \rangle\rangle X\top.$$

Coalition C has a strategy harmful to H .

In the robot model:

$$(M_{rs}, s_0) \models \text{Benef}(\{1, 2\}, \{1, 2\}) \wedge \text{Danger}(\{1, 2\}, \{0\}).$$

We axiomatize the until-free, or safety, fragment:

$$\varphi ::= p \mid \neg\varphi \mid \varphi \wedge \psi \mid \langle\!\langle C:B,H \rangle\!\rangle X\varphi \mid \langle\!\langle C:B,H \rangle\!\rangle G\varphi.$$

Inference rules

$$\frac{\varphi, \quad \varphi \rightarrow \psi}{\psi} \quad (\text{MP}) \qquad \frac{\varphi \leftrightarrow \psi}{\langle\!\langle C:B,H \rangle\!\rangle X\varphi \leftrightarrow \langle\!\langle C:B,H \rangle\!\rangle X\psi} \quad (\text{RE}) \qquad \frac{\varphi}{\langle\!\langle C \rangle\!\rangle G\varphi} \quad (\text{Gen})$$

Axiom schemata

All tautologies of propositional logic	(TAUT)
$\neg \langle\langle C:B, H \rangle\rangle X \perp$	(NAAA)
$\langle\langle C:B, H \rangle\rangle X(\varphi \wedge \psi) \rightarrow \langle\langle C:B, H \rangle\rangle X\varphi$	(MG)
$\langle\langle C:B, H \rangle\rangle X\varphi \rightarrow \langle\langle C':B', H' \rangle\rangle X\varphi$ for $C \subseteq C', B \supseteq B', H \supseteq H'$	(MC)
$\langle\langle \emptyset \rangle\rangle X \top$	(NEI)
$(\langle\langle C:B, H \rangle\rangle X\varphi \wedge \langle\langle C':B', H' \rangle\rangle X\psi) \rightarrow \langle\langle C \cup C':B \cup B', H \cup H' \rangle\rangle X(\varphi \wedge \psi)$ for $C \cap C' = \emptyset$	(Sup)
$\langle\langle C:B, H \rangle\rangle X(\varphi \vee \psi) \rightarrow (\langle\langle C:B, H \rangle\rangle X\varphi \vee \langle\langle \text{AGT}:B, H \rangle\rangle X\psi)$	(Cro)
$\langle\langle C:B, H \rangle\rangle G\varphi \leftrightarrow \langle\langle C:B, H \rangle\rangle X(\varphi \wedge \langle\langle C:B, H \rangle\rangle G\varphi)$	(FP _G)
$\langle\langle \emptyset \rangle\rangle G(\psi \rightarrow \langle\langle C:B, H \rangle\rangle X(\varphi \wedge \psi)) \rightarrow \langle\langle \emptyset \rangle\rangle G(\psi \rightarrow \langle\langle C:B, H \rangle\rangle G\varphi)$	(GFP _G)
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Theorem

For every $\varphi \in \mathcal{L}_{\text{WA-ATL-U}}$:

φ is valid over CGMRPs iff $\vdash \varphi$.

- The proof uses canonical models built under a formula context.
- The states in the canonical models are consist of (*finite*) maximally consistent sets of formulas in the formula context, and the labels of benefited and harmed groups.
- Stability of preferences keeps those labels coherent over time.

Theorem

For every $\varphi \in \mathcal{L}_{\text{WA-ATL}}$:

φ is satisfiable iff φ is satisfiable on a pointed tree-like CGMRP.

- Tree-like models have a unique root, unique predecessors, and no cycles.
- This lets us localize welfare labels along generated computations.

Embedding into Standard ATL

Extend the atomic vocabulary with:

$$\heartsuit_i \quad \text{and} \quad \spadesuit_i$$

read as “agent i is benefited” and “agent i is harmed”.

Translation:

$$\begin{aligned} tr(\langle\langle C:B,H \rangle\rangle X\varphi) &= \langle\langle C \rangle\rangle X(\heartsuit B \wedge \spadesuit H \wedge tr(\varphi)), \\ tr(\langle\langle C:B,H \rangle\rangle G\varphi) &= \langle\langle C \rangle\rangle G(\heartsuit B \wedge \spadesuit H \wedge tr(\varphi)), \\ tr(\langle\langle C:B,H \rangle\rangle (\varphi U \psi)) &= \langle\langle C \rangle\rangle ((\heartsuit B \wedge \spadesuit H \wedge tr(\varphi)) U (\heartsuit B \wedge \spadesuit H \wedge tr(\psi))), \end{aligned}$$

where

$$\heartsuit B = \bigwedge_{i \in B} \heartsuit_i, \quad \spadesuit H = \bigwedge_{i \in H} \spadesuit_i.$$

Embedding Theorem and Complexity

Polynomial embedding

Satisfiability for WA–ATL reduces polynomially to satisfiability for standard ATL, with additional constraints ensuring:

best labels exist, worst labels exist, and no state is both for the same agent.

Complexity

Language	Target	Satisfiability
WA–ATL	ATL	Exptime-complete
WA–CL	CL	Pspace-complete

The bounds are tight because ATL and CL are special cases.

Summary and Outlook

- Strategic agency can be welfare-affecting: agents can benefit or endanger others through their strategies.
- WA–ATL internalizes this idea in the object language:

$$\langle\langle C:B, H \rangle\rangle\gamma.$$

- Potential benefactor and danger become definable notions.
- The logic remains technically well-behaved:
 - complete axiomatization for the safety fragment;
 - polynomial embedding into standard ATL/CL with welfare-label atoms;
 - tight complexity bounds.

Next steps

Full axiomatization with until operators; probabilistic risk-based danger.

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Thank you

Questions?